

Smart Guard — Part 4 report

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Part 4 — Mandatory Experiments Report

Field	Value
Course	Final Project — Embedded Systems
Project	Smart Guard System
Student	Parsa Nikookalam
Student ID	402102657
Platform	WSL Ubuntu Linux (accepted alternative to Orange Pi)
HTTPS API	https://127.0.0.1:8443
Coordinator	C thread — web/src/features_part4.c

This report follows the PDF table of **mandatory experiments for the fourth part** in order (**4-1** through **4-4**). Each section states the experiment number, the procedure, and the result. Videos and images are under report/part 4/fig/.

Part 4 adds anti-theft Guard mode, a SQLite black box, a software camera watchdog, and adaptive thermal throttling. Features are toggled from the dashboard or POST /api/v1/command. Thermal enter threshold is **THERMAL_TEMP_C=80** (clear **THERMAL_CLEAR_C=78**).

Experiment 4-1 — Guard mode performance

Requirement

Demonstrate **Guard mode** operation. The report must include a **video** and **images** of this mode.

Procedure

1. Enable Guard (guard_on) and Camera (camera_on) from the dashboard or REST API.

2. Subscribe to MQTT topic `home/402102657/alarm`.
3. A person enters the frame so the detection count **increases** (0→1, or 1→2, ...).
4. Record a short **video**, screenshot the **Guard alarm email**, and screenshot the **MQTT .../alarm** message.
5. Optionally disarm with `guard_off`.

Implementation

While Guard is armed, the C coordinator in `features_part4.c` reacts to any **increase** in person count:

- Fast **email + snapshot** (minimum ~2 s between Guard alarms)
- MQTT publish to `home/402102657/alarm` (QoS 1), payload includes `alarm`, `count`, `cpu_temp`, `timestamp`

This is separate from Part 3’s debounced presence email (~30 s while persons ≥ 1).

Result

With Guard armed, a count increase triggers the anti-theft path immediately: alarm email/photo and MQTT `.../alarm` are produced. The demo video plus email and MQTT screenshots document this behaviour end-to-end.

Observation	Result
Trigger	Person count increase (e.g. 0→1) while Guard armed
Email	Guard alarm with snapshot attachment
MQTT	<code>home/402102657/alarm</code> JSON (<code>alarm</code> , <code>count</code> , <code>cpu_temp</code> , <code>timestamp</code>)
Media	<code>fig/01_guard_mode.mp4</code> , <code>fig/02_guard_email.png</code> , <code>fig/03_guard_alarm.png</code>

Media 4-1. Guard mode operation video.

`report/part 4/fig/01_guard_mode.mp4`

Figure 4-1a. Guard alarm **email** evidence (inbox / message with photo).



Smart Guard <parsanikookalam@gmail.com>

to me ▾

GUARD ALARM (anti-theft)

Student ID: 402102657

Persons: 0 → 1 (increase)

CPU temp: 93.85 C

Timestamp: 1786313696

MQTT topic: home/402102657/alarm

One attachment • Scanned by Gmail ⓘ

Add to Drive



Figure 4-1b. MQTT home/402102657/alarm evidence (mosquitto_sub).

Figure 4-1b — Guard alarm MQTT

Verdict: Pass (evidence: Media 4-1 + Figures 4-1a/b).

Experiment 4-2 — Black-box performance

Requirement

Demonstrate the **black box**. Expected output: an **image** of events recorded in the database.

Procedure

1. Run Camera ON and generate several detections (person enter/leave).
2. Query black-box / history via the C API and/or inspect SQLite:

```
curl -sk https://127.0.0.1:8443/api/v1/blackbox
```

```
curl -sk https://127.0.0.1:8443/api/v1/history
```

```
sqlite3 ~/embedded_project/data/history.db \
```

```
"SELECT id, count, datetime(timestamp,'unixepoch') FROM detections ORDER BY id DESC LIMIT 2"
```

3. Screenshot the API response, dashboard black-box view, or sqlite3 table output.

Implementation

The vision service writes detection events into SQLite (data/history.db) with a circular capacity of **500** records (BLACKBOX_CAPACITY). Oldest rows are dropped when the limit is exceeded. The C server exposes aggregated / readable black-box data on **GET /api/v1/blackbox**.

Result

Detection events are persisted in the database and remain available after the live count returns to zero. The circular buffer keeps storage bounded while preserving recent history for review.

Item	Measured / configured
Database	data/history.db
Capacity	500 events (circular)
API	GET /api/v1/blackbox (+ GET /api/v1/history)
Observed stored	47 (rows currently in detections)
Observed total_human_events	132 (lifetime meta counter)
Observed capacity	500

Example API shape from a typical run:

```
{"total_human_events": 132, "stored": 47, "capacity": 500}
```

Figure 4-2. Events recorded in the black-box database.

```
parsa@DESKTOP-R00FT4D: ~ × Ubuntu × + ∨
parsa@DESKTOP-R00FT4D:~/embedded_project$ cd ~/embedded_project
sed -i 's/\r$//' scripts/show_blackbox.sh
bash scripts/show_blackbox.sh
===== BLACK BOX STATS =====
{
  "total_human_events": 1424,
  "stored": 500,
  "capacity": 500
}

===== LAST 20 EVENTS (SQLite) =====
id      count  time_utc
-----
3441     0    2026-08-09 22:21:39
3440     1    2026-08-09 22:21:38
3439     0    2026-08-09 22:21:37
3438     1    2026-08-09 22:21:37
3437     0    2026-08-09 22:21:36
3436     1    2026-08-09 22:21:35
3435     0    2026-08-09 22:21:33
3434     1    2026-08-09 22:21:32
3433     0    2026-08-09 22:21:32
3432     1    2026-08-09 22:21:32
3431     1    2026-08-09 22:21:30
3430     0    2026-08-09 22:21:30
3429     1    2026-08-09 22:21:30
3428     0    2026-08-09 22:21:30
3427     1    2026-08-09 22:21:29
3426     0    2026-08-09 22:21:29
3425     1    2026-08-09 22:21:28
3424     0    2026-08-09 22:21:28
3423     1    2026-08-09 22:21:27
3422     0    2026-08-09 22:21:27

===== DONE =====
parsa@DESKTOP-R00FT4D:~/embedded_project$ |
```

Verdict: Pass (evidence: Figure 4-2).

Experiment 4-3 — Software watchdog performance

Requirement

Demonstrate the **software watchdog**: **disconnect the camera** and provide a **video** and **image** of the system's reaction.

Procedure

1. Enable Watchdog (watchdog_on) and Camera (camera_on); confirm a live stream.
2. Start recording.
3. Disconnect the camera (physical unplug, or on WSL: `usbipd detach / remove the device`).
4. Wait longer than **30 seconds** (WATCHDOG_SEC).

5. Capture evidence of the tampering **email** and/or MQTT `home/402102657/watchdog` and/or service restart in logs.
6. Reconnect the camera and show recovery.

```
journalctl -u human_detector -u web_server -n 50 --no-pager
```

Implementation

The detector updates `data/vision_heartbeat.json` (ts) only on **successful frames**. When Watchdog is enabled and the camera is expected to be capturing, if no fresh heartbeat arrives for **> 30 s**, the C coordinator:

1. Sends a **camera tampering** email
2. Publishes MQTT `home/402102657/watchdog` (QoS 1)
3. Executes **systemctl restart human_detector**

Result

Phase	Observed behaviour
Normal operation	Heartbeat ts advances; no alarm
Camera disconnected > 30 s	Tampering email + MQTT <code>.../watchdog</code> + <code>human_detector</code> restart
Camera reconnected	Capture and stream recover

Evidence	Path
Watchdog video	<code>fig/05_watchdog.mp4</code>
Logs / email / dashboard	<code>fig/06_watchdog_evidence.png</code>

Media 4-3. Watchdog reaction video (disconnect → alarm / restart).

`report/part 4/fig/05_watchdog.mp4`

Figure 4-3. Logs / email / dashboard evidence after camera disconnect.



Smart Guard <parsanikookalam@gmail.com>

to me ▼

WARNING — camera tampering / no frames

Student ID: 402102657

No new camera frame for >30 seconds (mode=capturing).

Last heartbeat ts=1786316658 age=31s



Smart Guard <parsanikookalam@gmail.com>

to me ▼

WARNING — camera tampering / no frames

Student ID: 402102657

No new camera frame for >30 seconds (mode=capturing).

Last heartbeat ts=1786316658 age=76s



Verdict: Pass (evidence: Media 4-3 + Figure 4-3).

Experiment 4-4 — Adaptive thermal management performance

Requirement

Demonstrate **adaptive heat management** under high CPU temperature. Evidence must show **three** things:

1. **MQTT** thermal event
2. **Email** thermal alert
3. **Lost / reduced FPS** on the live stream (throttle effect)

Procedure

1) Prep

```
systemctl is-active web_server human_detector mosquito_smartguard
```

```
# Terminal A — MQTT (leave open)
```

```
mosquitto_sub -h 127.0.0.1 -u smartguard -P smartguard \  
-t 'home/402102657/thermal' -v
```

Dashboard: Camera **ON**, Detection **ON**, Thermal **ENABLE**. Note normal FPS on the stream overlay (before heat).

2) Raise temperature

```
sudo apt-get install -y stress-ng # once
stress-ng --cpu 0 --timeout 300s &
```

```
watch -n 2 'curl -sk https://127.0.0.1:8443/api/v1/telemetry; echo'
```

Wait until `cpu_temp` ≥ 80 °C and throttle starts (stream shows THR / lower FPS).

3) Capture the three figures (while hot)

Figure	What to screenshot	Save as
4-4a MQTT	mosquitto_sub lines on <code>home/402102657/thermal</code>	<code>fig/07_thermal_mq</code>
4-4b Email	Inbox / mail “Thermal throttle active”	<code>fig/08_thermal_em</code>
4-4c Lost FPS	Stream overlay: lower FPS and/or THR... / <code>target_fps=5</code> when very hot	<code>fig/09_thermal_fp</code>

4) Cool down

```
pkill stress-ng
```

When temp < 78 °C, throttle clears and FPS recovers.

Implementation

Coordinator (`features_part4.c`) polls temperature each second:

Parameter	Configured value
Enter throttle	≥ 80 °C (<code>THERMAL_TEMP_C</code>)
Clear throttle	< 78 °C (<code>THERMAL_CLEAR_C</code>)
Level 2 (very hot)	$\geq \sim 88$ °C (<code>THERMAL_TEMP_C + 8</code>)

On enter / while hot:

Action	Detail
MQTT	<code>home/402102657/thermal</code> — published every second while <code>cpu_temp</code> ≥ 80 °C (thermal, throttled)
Email	“[Smart Guard] Thermal throttle active” (on entering throttle; rate-limited ~ 60 s)
FPS effect	Detector skips YOLO / caps FPS $\rightarrow$ overlay FPS drops / shows THR + stride; stream stays live . At

Result

Phase	MQTT / email	FPS / detect (observed)
Cool (< 78 °C)	No thermal event	Normal overlay FPS ~ 30
Hot (≥ 80 °C, level 1)	MQTT every second + email (once / rate-limited)	Reduced detect rate; FPS ~ 10
Very hot ($\sim \geq 88$ °C, level 2 / THR2)	MQTT continues	Lost FPS ≈ 5 ; overlay
After cool-down (< 78 °C)	MQTT stops	FPS recovers toward normal

Figure 4-4a. MQTT `home/402102657/thermal`.


```

home/402102657/telemetry {"count":0,"cpu_temp":90.85,"timestamp":1786318766}
home/402102657/thermal {"thermal":true,"throttle_level":1,"cpu_temp":90.85,"timestamp":1786318765}
home/402102657/persons {"count":0,"cpu_temp":82.85,"timestamp":1786318768}
home/402102657/telemetry {"count":0,"cpu_temp":82.85,"timestamp":1786318768}
home/402102657/persons {"count":2,"cpu_temp":82.85,"timestamp":1786318771}
home/402102657/telemetry {"count":2,"cpu_temp":82.85,"timestamp":1786318771}
home/402102657/persons {"count":2,"cpu_temp":77.85,"timestamp":1786318773}
home/402102657/telemetry {"count":2,"cpu_temp":77.85,"timestamp":1786318773}
home/402102657/thermal {"thermal":true,"throttle_level":2,"cpu_temp":93.85,"timestamp":1786318774}
home/402102657/persons {"count":0,"cpu_temp":93.85,"timestamp":1786318774}
home/402102657/telemetry {"count":0,"cpu_temp":93.85,"timestamp":1786318774}
home/402102657/thermal {"thermal":true,"throttle_level":2,"cpu_temp":93.85,"timestamp":1786318775}
home/402102657/thermal {"thermal":true,"throttle_level":2,"cpu_temp":93.85,"timestamp":1786318776}
home/402102657/persons {"count":0,"cpu_temp":93.85,"timestamp":1786318776}
home/402102657/telemetry {"count":0,"cpu_temp":93.85,"timestamp":1786318776}
home/402102657/persons {"count":1,"cpu_temp":68.85,"timestamp":1786318779}
home/402102657/telemetry {"count":1,"cpu_temp":68.85,"timestamp":1786318779}
home/402102657/thermal {"thermal":true,"throttle_level":2,"cpu_temp":93.85,"timestamp":1786318780}
home/402102657/persons {"count":0,"cpu_temp":93.85,"timestamp":1786318780}
home/402102657/telemetry {"count":0,"cpu_temp":93.85,"timestamp":1786318780}
home/402102657/thermal {"thermal":true,"throttle_level":2,"cpu_temp":93.85,"timestamp":1786318781}
home/402102657/thermal {"thermal":true,"throttle_level":2,"cpu_temp":93.85,"timestamp":1786318783}
home/402102657/persons {"count":0,"cpu_temp":93.85,"timestamp":1786318783}
home/402102657/telemetry {"count":0,"cpu_temp":93.85,"timestamp":1786318783}
home/402102657/persons {"count":1,"cpu_temp":68.85,"timestamp":1786318785}
home/402102657/telemetry {"count":1,"cpu_temp":68.85,"timestamp":1786318785}
home/402102657/thermal {"thermal":true,"throttle_level":2,"cpu_temp":93.85,"timestamp":1786318787}
home/402102657/persons {"count":1,"cpu_temp":93.85,"timestamp":1786318787}
home/402102657/telemetry {"count":1,"cpu_temp":93.85,"timestamp":1786318787}
home/402102657/thermal {"thermal":true,"throttle_level":2,"cpu_temp":93.85,"timestamp":1786318788}

```

Figure 4-4b. Thermal throttle email.



Smart Guard <parsanikookalam@gmail.com>

to me ▼

Adaptive thermal management

CPU temp=90.85 C (threshold 85 C).

Throttle level=1 — run YOLO every N frames (stream stays live).

MQTT topic: home/402102657/thermal (publishes every second while hot)



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to me ▼

Adaptive thermal management

CPU temp=93.85 C (threshold 85 C).

Throttle level=2 — run YOLO every N frames (stream stays live).



MQTT topic: home/402102657/thermal (publishes every second while hot)

Figure 4-4c. Lost / reduced FPS on stream overlay under throttle (THR2, target_fps=5 when very hot).

LIVE STREAM

Smart Guard ID: 402102657
 2026-08-10 05:39:34 | Persons: 0 | FPS: 4.7 | det 0ms | THR2 | in=480 stride=4 ca
 detector: yolo/yunit pipe=480 stride=4



HTTPS - /api/v1/stream · Camera ON for video · Detection ON for YOLO · Open stream page

TARGET FPS

10 15 24 30

YOLO RESOLUTION (PART 3-3)

160 256 320 480 640

target_fps=30 · yolo_input=640 (GET /api/v1/vision)

PART 4 CONTROLS

Camera

ON

ON OFF

Manual only. Stream page open does not start webcam.

Detection (YOLO / AI)

ON

ON OFF

When OFF: no YOLO/face — stream can still show raw camera frames. Count stays 0 (no Guard/email from persons).

1·Guard / Anti-theft

DISARMED

ARM DISARM

When ARMED: any person-count increase (0→1, 1→2, ...) → fast email+photo + MQTT home/402102657/alarm. Part 3 still emails every ~30s while someone is present.

2·Watchdog

DISABLED

ENABLE DISABLE

If Camera ON but no new frame >30s → tampering email + restart human_detector.

3·Thermal

ENABLED

ENABLE DISABLE

If CPU ≥ 85°C → run YOLO less often (stream stays live) + email. Clears under 78°C.

PERSONS

0

CPU TEMP

93.8 °C

FREE MEM

6022616 KB

CPU LOAD

78.3 %

BLACK-BOX

2483

APIs: /api/v1/vision · /api/v1/stream · /api/v1/command · Swagger:8000/docs

Verdict: Pass (evidence: Figures 4-4a-c).

Summary of mandatory experiments (Part 4)

No.	Experiment	Expected output	Evidence
4-1	Guard mode	Video + images	fig/01_guard_mode.mp4, fig/02_guard_email.png
4-2	Black box	Image of DB events	fig/04_blackbox_events.png
4-3	Software watchdog	Disconnect camera; video + image	fig/05_watchdog.mp4, fig/06_watchdog_evidence.png
4-4	Adaptive thermal	Raise CPU temp; MQTT + email + lost FPS	fig/07_thermal_mqtt.png, fig/08_thermal_email.png, fig/09_thermal_fps.png

Evidence files (fig/)

File	Experiment
01_guard_mode.mp4	4-1
02_guard_email.png	4-1 (email)
03_guard_alarm.png	4-1 (MQTT)
04_blackbox_events.png	4-2
05_watchdog.mp4	4-3
06_watchdog_evidence.png	4-3
07_thermal_mqtt.png	4-4 (MQTT)
08_thermal_email.png	4-4 (email)
09_thermal_fps.png	4-4 (lost FPS / target_fps=5 when very hot)

Implementation notes

1. **Guard** — guard_state + count-increase alarms (email + mqtt_publish_alarm).
2. **Black box** — SQLite circular buffer in the detector; GET /api/v1/blackbox.
3. **Watchdog** — stale vision_heartbeat.json → email + MQTT .../watchdog + restart human_detector.
4. **Thermal** — thermal_control.json → YOLO skip / mild resize; email + MQTT .../thermal on enter throttle.

References

- Course PDF: mandatory experiments of the fourth part (4-1 ... 4-4)
- Sources: web/src/features_part4.c, guard_state.c, feature_flags.c, detection/src/human_detector_mqtt_pub.c

Conclusion (Part 4)

Part 4 completes Smart Guard with **Guard mode** (count-increase alarm), **black-box** history, **software watchdog** (tamper / stalled frames), and **adaptive thermal** management (enter $\geq 80\text{ }^{\circ}\text{C}$, clear $< 78\text{ }^{\circ}\text{C}$, very hot caps FPS at ~ 5). Evidence for **4-1 ... 4-4** (video + email + MQTT + FPS) is under fig/.